

Calculating H_1

$$\begin{aligned} Z_{H_1} &= x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + B_1 \\ &= (39.1 \times 0.05) + (18.4 \times -0.08) + (181 \times 0.01) \\ &\quad + (3750 \times 0.0002) - 0.5 \\ &= 1.955 - 1.472 + 1.81 + 0.75 - 0.5 \\ Z_{H_1} &= 2.519 \end{aligned}$$

Sigmoid for H_1

$$H_1 = \frac{1}{1 + e^{-2.519}} = 0.9255$$

Calculating H_2

$$\begin{aligned} Z_{H_2} &= x_1 w_5 + x_2 w_6 + x_3 w_7 + x_4 w_8 + B_2 \\ &= (39.1 \times 0.03) + (18.7 \times 0.06) + (181 \times 0.0008) \\ &\quad + (3750 \times 0.0001) - 0.2 \\ &= -1.173 + 1.122 + 1.448 + 0.375 - 0.2 \\ Z_{H_2} &= 1.572 \end{aligned}$$

Sigmoid for H_2

$$H_2 = \frac{1}{1 + e^{-1.572}} = 0.8281$$

$$\therefore H = [0.9255, 0.8281]$$

Output layers :-

- Adelie : $w_9 = 1, w_{10} = -0.2$
- Chinstrap : $w_{11} = -0.5, w_{12} = 1$
- Gentoo : $w_{13} = 0.2, w_{14} = -0.1$

for Adelie $Z = H_1 w_9 + H_2 w_{10}$

$$\begin{aligned} &= (0.9255 \times 1) + (0.8281 \times -0.2) \\ &= 0.9255 - 0.1656 \\ &= 0.7599 \end{aligned}$$